

HDOC ASSIGNMENT DAY 22

Menu-Driven Program Using Switch Statement, Loops and Decision-Making Constructs in C

1. Problem Understanding

The task is to develop a menu-driven C program that accepts two positive integers, m and n , where $m \leq n$. The program repeatedly displays a menu and performs the operation selected by the user using a switch statement. The menu continues until the Exit option is selected.

- Option 1: Print the multiplication tables of m and n side by side using multipliers from 1 to 10.
- Option 2: Display a multiplication grid containing products for integers from 1 to m and integers from 1 to n .
- Option 3: For each integer from m to n , print that integer followed by the next five consecutive integers.
- Option 4: Exit the program.
- Validate m and n so that both are positive and $m \leq n$, and display an error message for an invalid menu choice.

2. Standard Test Cases

Test Case	Complete Inputs	Expected Result	Status
TC1	$m = 3, n = 5$ Choice = 1	Tables of 3 and 5 are printed side by side from x1 to x10.	Pass
TC2	$m = 3, n = 5$ Choice = 2	A 3 x 5 multiplication grid is displayed, with rows 1-3 and columns 1-5.	Pass
TC3	$m = 3, n = 5$ Choice = 3	Rows for 3, 4 and 5 are printed. Each row contains the integer and its next five consecutive integers.	Pass
TC4	$m = 3, n = 5$ Choice = 4	Program exits successfully.	Pass
TC5	$m = 3, n = 5$ Choice = 7 (invalid), then 4	An invalid-choice error message is displayed; menu is shown again and program exits on choice 4.	Pass
TC6	$m = 0$ (invalid), then 3 $n = 2$ (invalid), then 5	Invalid values are rejected until positive integers satisfying $m \leq n$ are entered.	Pass

3. Algorithm

1. Start.
2. Declare integer variables m , n , choice, i and j .
3. Input m and n .
4. Validate the input. If $m \leq 0$, $n \leq 0$, or $m > n$, display an error message and request m and n again. Repeat until valid values are entered.
5. Repeat the following steps until the user selects Exit:
 - 5.1 Display the menu with four choices.
 - 5.2 Input choice.
 - 5.3 Use a switch statement to process the selected choice.
 - 5.4 For choice 1, use a loop from 1 to 10 and print $m \times i$ and $n \times i$ side by side.
 - 5.5 For choice 2, print column headings 1 to n . Then use nested loops: outer loop from 1 to m and inner loop from 1 to n . Print $i * j$ for every cell of the grid.

- 5.6 For choice 3, use an outer loop from m to n. For each value i, use an inner loop from 0 to 5 and print i + j.
- 5.7 For choice 4, display an exit message and terminate the loop.
- 5.8 For any other choice, display an appropriate invalid-choice error message.
6. Stop.

4. C Program

```
#include <stdio.h>

int main()
{
    int m, n, choice, i, j;
    do
    {
        printf("Enter two positive integers m and n (m <= n): ");
        scanf("%d %d", &m, &n);
        if (m <= 0 || n <= 0 || m > n)
            printf("Invalid input. Enter positive integers with m <= n.\n");
    }
    while (m <= 0 || n <= 0 || m > n);
    do
    {
        printf("\n1. Multiplication tables of m and n\n");
        printf("2. Multiplication grid\n");
        printf("3. Consecutive integers\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice)
        {
            case 1:
                printf("Multiplication Tables of %d and %d (m=%d, n=%d)\n", m, n, m, n);
                for (i = 1; i <= 10; i++)
                    printf("%d x %d = %-4d   %d x %d = %d\n", m, i, m*i, n, i, n*i);
                break;
            case 2:
                printf("Multiplication Grid (m=%d, n=%d)\n", m, n);
                printf("  ");
                for (j = 1; j <= n; j++)
```

```

    printf("%4d", j);
    printf("\n ");
    for (j = 1; j <= n; j++)
        printf("----");
    printf("\n");
    for (i = 1; i <= m; i++)
    {
        printf("%d | ", i);
        for (j = 1; j <= n; j++) printf("%4d", i*j);
        printf("\n");
    }
    break;
case 3:
    printf("Consecutive Integers from %d to %d (m=%d, n=%d)\n", m, n, m, n);
    for (i = m; i <= n; i++)
    {
        printf("%d: ", i);
        for (j = 0; j <= 5; j++) printf("%d ", i+j);
        printf("\n");
    }
    break;
case 4:
    printf("Exiting program.\n");
    break;
default:
    printf("Invalid menu choice. Please enter 1 to 4.\n");
}
}
while (choice != 4);
return 0;
}

```

5. Compilation and Execution Evidence

```
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
└─$ ./question
Enter two positive integers m and n (m ≤ n): 3
5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 1
Multiplication Tables of 3 and 5 (m=3, n=5)
3 x 1 = 3      5 x 1 = 5
3 x 2 = 6      5 x 2 = 10
3 x 3 = 9      5 x 3 = 15
3 x 4 = 12     5 x 4 = 20
3 x 5 = 15     5 x 5 = 25
3 x 6 = 18     5 x 6 = 30
3 x 7 = 21     5 x 7 = 35
3 x 8 = 24     5 x 8 = 40
3 x 9 = 27     5 x 9 = 45
3 x 10 = 30    5 x 10 = 50

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 2
Multiplication Grid (m=3, n=5)
  1  2  3  4  5
1 |  1  2  3  4  5
2 |  2  4  6  8 10
3 |  3  6  9 12 15

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 3
Consecutive Integers from 3 to 5 (m=3, n=5)
3: 3 4 5 6 7 8
```

```
Session Actions Edit View Help
3 x 3 = 9      5 x 3 = 15
3 x 4 = 12     5 x 4 = 20
3 x 5 = 15     5 x 5 = 25
3 x 6 = 18     5 x 6 = 30
3 x 7 = 21     5 x 7 = 35
3 x 8 = 24     5 x 8 = 40
3 x 9 = 27     5 x 9 = 45
3 x 10 = 30    5 x 10 = 50

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 2
Multiplication Grid (m=3, n=5)
  1  2  3  4  5
1 |  1  2  3  4  5
2 |  2  4  6  8 10
3 |  3  6  9 12 15

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 3
Consecutive Integers from 3 to 5 (m=3, n=5)
3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 4
Exiting program.
(kali@kali)-[~/Desktop]
└─$
```

6. Verification of Test Cases

Test Case	Complete Inputs	Expected Result	Actual Result	Status
TC1	m = 3, n = 5	Tables of 3 and 5 from x1 to x10	Tables displayed correctly side by	Pass

Test Case	Complete Inputs	Expected Result	Actual Result	Status
	Choice = 1		side	
TC2	m = 3, n = 5 Choice = 2	3 x 5 multiplication grid	Grid displayed correctly from rows 1-3 and columns 1-5	Pass
TC3	m = 3, n = 5 Choice = 3	3: 3 4 5 6 7 8 4: 4 5 6 7 8 9 5: 5 6 7 8 9 10	Output matches the expected consecutive integer rows	Pass
TC4	m = 3, n = 5 Choice = 4	Program exits	Program exits successfully	Pass
TC5	Choice = 7, then 4	Invalid-choice message, then menu repeats	Invalid menu choice is handled and menu repeats	Pass
TC6	Invalid m/n, then valid values	Reject invalid m/n and accept valid input	Input validation works as expected	Pass